CS & DA

Probability and Statistics

Probability

Lecture No. 03





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Recap of previous lecture



Topic

Probability (Part 2)

Topics to be Covered



Topic

BASICS of PROBABILITY (Part-03)

(*) Fav Cases :→ Fav = { which is Required & should be assumed as Fav }

(*) Revision : = { Desired outcome " " " " }

Operation	P ∈ C	Prob	ME	Ind
Either or	Add	∪ use Addition Th	$P(A \cup B) = P(A) + P(B)$	$P(A \cup B) = P(A) + P(B) - P(A) \cdot P(B)$
AND	Multiply	∩ use Multi Th	$P(A \cap B) = 0$	$P(A \cap B) = P(A) \cdot P(B)$

(*) (1) Three Dice are thrown simultaneously
(2) A Die is thrown three times

Both have same S space

$S = \{(111)(112)\ldots(116)\ (211)(212)\ldots(666)\} \Rightarrow n(S) = 6 \times 6 \times 6 = 216$ Triplets

PQ.2: A coin is tossed 6 times then write it's S.Space.

$$S=\{(HHHHHH), (HHHHHT), (HHHHTT), (HHHTTT), (HHTTTT), (HTTTTT), (TTTTTT)\}$$

$${}^6C_6=1 \quad {}^6C_5=6 \quad {}^6C_4=15 \quad {}^6C_3=20 \quad {}^6C_2=15 \quad {}^6C_1=6 \quad {}^6C_0=1$$

$$n(S)=\frac{2}{C_1}\times\frac{2}{C_2}\times\frac{2}{C_3}\times\frac{2}{C_4}\times\frac{2}{C_5}\times\frac{2}{C_6}=2^6=64 \text{ ordered 6 in tuples.}$$

$(HHHHHH) \longrightarrow \frac{6!}{6!}=1={}^6C_6$

$(HHHHHT) \longrightarrow \frac{6!}{5!}=6={}^6C_5$

$(HHHHTT) \longrightarrow \frac{6!}{4!2!}=15={}^6C_4$

$(HHHTTT) \longrightarrow \frac{6!}{3!3!}=20={}^6C_3$

$(HHTTTT) \longrightarrow \frac{6!}{2!4!}=15={}^6C_2$

$(HTTTTT) \longrightarrow \frac{6!}{5!}=6={}^6C_1$

$(TTTTTT) \longrightarrow \frac{6!}{6!}=1={}^6C_0$

In above Exp; $P(H)=P(T)=\frac{1}{2}$ & Individual tosses are Independent

① Find the prob that all the outcomes are identical?

App I: fav elements $=\{(HHHHHH),(TTTTTT)\}$

$=2$

So Req Prob $=\frac{\text{fav}}{\text{Total}}=\frac{2}{64}=\frac{1}{32}$

App II: Req Prob $=P[(HHHHHH)\text{ or }(TTTTTT)]$ (ME)

$=\left(\frac{1}{2}\right)^6+\left(\frac{1}{2}\right)^6=\left(\frac{1}{2}\right)^5$

App III → Prob $=\frac{\text{fav}}{T}=\frac{{}^6C_6+{}^6C_6}{2^6}=\frac{2}{64}$

② Find the prob that H & T appears alternately?

App I: fav Cases $=\{(HTHTHT),(THTHTH)\}$

$=2$

Req Prob $=\frac{\text{fav}}{\text{Total}}=\frac{2}{64}=\frac{1}{32}$

App III: Req Prob $=P[(HTHTHT)\text{ or }(THTHTH)]$ (ME)

$=\left(\frac{1}{2}\right)^6+\left(\frac{1}{2}\right)^6=\left(\frac{1}{2}\right)^5=\frac{1}{32}$

(3) Find the prob that Both H & T appears at least once ?

App I: unfav outcomes $= \{(HHHHHH), (TTTTTT)\}$

$= 2$

$\therefore$ fav outcomes $= 64 - 2 = 62$

Req Prob $= \frac{\text{fav}}{\text{Total}} = \frac{62}{64}$

App III: Req Prob $= 1 - P(\text{all H or all T})$

$= 1 - \frac{2}{64} = \frac{62}{64}$

(4) Find the prob that H appears at least once ?

App I: unfav outcomes $= \{(TTTTTT)\} = 1$

$\therefore$ fav outcome $= 64 - 1 = 63$

Req Prob $= \frac{\text{fav}}{\text{Total}} = \frac{63}{64}$

App III: $P(\text{at least one H}) = 1 - P(\text{No Head})$

$= 1 - P(\text{all T})$

$= 1 - P(TTTTTT)$

$= 1 - \left(\frac{1}{2}\right)^6$

⑤ If 1st 2 outcomes are H, H then

Find the prob of occurring T when Coin is tossed again

Sol:

Method-I: Req Prob = P(T in 3rd toss) = $\frac{1}{2}$

Method-II (App III): i.e.

Req Prob = P[(HH) T something occurs] (HH: given)

$= 1 \times 1 \times \frac{1}{2} \times 1 \times 1 \times 1 = \frac{1}{2} \neq \frac{32}{64}$

⑥ If 1st two outcomes are Both Heads then Find the prob of occurring Tail in Remaining tosses ?

App III

Req Prob = P[(HH) TTTT] (HH: given)

$= 1 \times 1 \times \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2}$

$= (1)^2 \times \left(\frac{1}{2}\right)^4 = \frac{1}{16} \neq \frac{4}{64}$

Note: whenever Question is Based on Conditional Prob, we should avoid to generalise our answer.

i.e. $\frac{32}{64} = \frac{1}{2}$ (✓) But $\frac{1}{2} = \frac{32}{64}$ ✗

(1) Find the prob that exactly 2 H appears ?

App I — fav cases = {(HHTTTT), (HTHTTT), (TTTHTH), ...} ☹

= Not easy to count

App II — fav cases = { eg (HTTHTT) ..... }

$$= \frac{6!}{2!\,4!} \simeq {}^{6}C_2 \times {}^{4}C_4 = 15$$

$$\text{Req Prob} = \frac{\text{fav}}{\text{Total}} = \frac{{}^{6}C_2}{2^6} = \frac{15}{64}$$

App III — use Binomial Dist → will be discussed later

(2) Find the prob that H & T appears equal number of times ?

App I → No easy

App II — fav cases = { H & T appears equal No of times }

= { exactly 3 H appears }

= { eg (HTHTTH) .... } $= {}^{6}C_3 \times {}^{3}C_3$

$$\text{Req Prob} = \frac{\text{fav}}{\text{Total}} = \frac{{}^{6}C_3}{2^6} = \frac{20}{64}$$

App III → using R-Distribution

⑨ Find the prob that, 1st two outcomes are Heads

Sol: App I fav cases = {(HH _ _ _ _ _ _) .....} = ?

App II fav cases = {eg (HH H/T H/T H/T H/T H/T) .....}
(3rd, 4th, 5th, 6th)

$= 1 \times 1 \times 2 \times 2 \times 2 \times 2 = 16$

So Req Prob $= \frac{\text{fav}}{\text{Total}} = \frac{16}{64} = \frac{1}{4}$

App III Req Prob $= P[(HH \text{ Something occurs})]$

$= \frac{1}{2} \times \frac{1}{2} \times 1 \times 1 \times 1 \times 1 = \frac{1}{4}$

⑩ Find the prob that (only) 1st two outcomes are Heads)

App I fav cases = {(HH TTTT)} = 1

Req Prob $= \frac{\text{fav}}{\text{Total}} = \frac{1}{64}$

App II → No Need

App III

Req Prob $= P[\text{only 1st two Head}] = P[HH\,TTTT]$

$= \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2}$

$= \left(\frac{1}{2}\right)^2 \times \left(\frac{1}{2}\right)^4 = \left(\frac{1}{2}\right)^6$

Explanation of 89 :

$$\text{Req Prob} = \frac{\text{fav}}{\text{Total}} = \frac{(HH\ HT\ HT\ HT\ HT\ HT)}{\text{Total}}$$

$$= \frac{1\times1\times2\times2\times2\times2}{64} = \frac{1}{2}\times\frac{1}{2}\times\frac{2}{2}\times\frac{2}{2}\times\frac{2}{2}\times\frac{2}{2}$$

App II

$$= \frac{1}{2}\times\frac{1}{2}\times1\times1\times1\times1 = P(HH \text{ something occurs})$$

= App III

$$64 = 2^6 = 2\times2\times2\times2\times2\times2$$

Diff b/n Q6 & Q10 -

(6) If 1st two outcomes are H, H then find the prob of occurring Tail in Remaining tosses

Condition

App III Req Prob = $P[\underset{\text{fix}}{(H\ H)}TTTT] = 1 \times 1 \times \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} = \frac{1}{16}$

(10) Find the Prob that only 1st two tosses produces Head

App III Req Prob = $P[HHTTTT] = \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} = \left(\frac{1}{2}\right)^6 = \frac{1}{64}$

HW8 A Coin is tossed 10 times then Find the Prob that

① Exactly 4 H appears = ? $= \frac{fav}{Total} = \frac{{}^{10}C_4 \times {}^{6}C_6}{2^{10}} = \frac{\frac{10!}{4!6!}}{1024} = \frac{210}{1024}$

② 4th Head appears in 9th toss = ?

App II: fav Cases = $\{eg\ (\underbrace{\text{exactly 3H}}_{\text{in 1st 8 tosses}}\ \ H\ (9^{th})\ \ H/T\ (10^{th}))\ \ldots\ldots\} = {}^{8}C_3 \times 1 \times 2$

Req Prob $= \frac{fav}{Total} = \frac{{}^{8}C_3 \times 1 \times 2}{2^{10}} = \frac{{}^{8}C_3}{2^8} \times \frac{1}{2} \times \frac{2}{2} = \frac{56}{512}$

App III:- Req Prob $= P\{\text{getting } 4^{th} \text{ H in } 9^{th} \text{ toss}\}$

$= P\{\text{exactly 3H in 1st 8 tosses}\} \times P(\text{H in } 9^{th} \text{ toss}) \times P(\text{Something occurs in } 10^{th} \text{ toss}) = \frac{{}^{8}C_3}{2^8} \times \frac{1}{2} \times 1$

Concept of ME and Independency in a Single Question →

eg Two persons fire at the target once then write it's S.Space

$S = \{$ None will hit , A hit & B missed , A missed & B hit , Both will hit $\}$

$= \{ \bar{A} \cap \bar{B} , A \cap \bar{B} , \bar{A} \cap B , A \cap B \} = \{ E_1, E_2, E_3, E_4 \}$

$\bar{A} \cap \bar{B} = E_1$, $A \cap \bar{B} = E_2$, $\bar{A} \cap B = E_3$, $A \cap B = E_4$

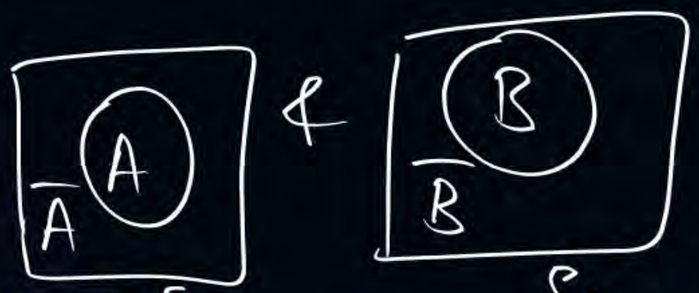
Here E_1, E_2, E_3, E_4 are ME Events :: these are the Elements of S.Sp.

while A & B are Ind ::

A & $\bar{B}$ " "

$\bar{A}$ & B " "

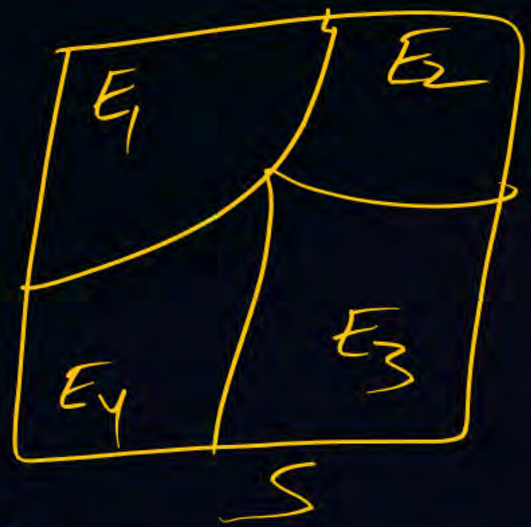
$\bar{A}$ & $\bar{B}$ " "



Here $(\bar{A} \cap \bar{B}) \cup (\bar{A} \cap B) \cup (A \cap \bar{B}) \cup (A \cap B) =$

Exactly one will hit

Hence $(\bar{A} \cap B) \cup (A \cap \bar{B}) \cup (\bar{A} \cap \bar{B}) \cup (A \cap B) = S$



At least one will hit

ie $E_1 \cup E_2 \cup E_3 \cup E_4 = S$

$P(E_1 \cup E_2 \cup E_3 \cup E_4) = P(S)$

$P(E_1) + P(E_2) + P(E_3) + P(E_4) = 1$

Here E_1, E_2, E_3, E_4 are ME events

⊛ $P(\text{at least one will hit}) = 1 - P(\text{None will hit})$

⊛ $P(E_4) = ? = P(A \cap B) = P(A) \cdot P(B)$

Q. There are two Gangsters MUNNA MOBILE (= A) & PAPPU PAZER (= B). They both fire at the Target once with prob of their hitting is $\frac{4}{5}$ & $\frac{3}{4}$ resp. then find the prob,

$P(A) = \frac{4}{5}$, $P(B) = \frac{3}{4}$

$P(\bar{A}) = \frac{1}{5}$, $P(\bar{B}) = \frac{1}{4}$

, A & B are Independent.

App III (1) Both will hit = ? $\Rightarrow P(A \cap B) = P(A) \cdot P(B) = \frac{4}{5} \times \frac{3}{4} = \frac{12}{20} = \frac{3}{5}$

App III (2) None will hit = ? $\Rightarrow P(\bar{A} \cap \bar{B}) = P(\bar{A}) \cdot P(\bar{B}) = \frac{1}{5} \times \frac{1}{4} = \frac{1}{20}$

App III (3) P(at least one will hit) = ? $= 1 - P(\text{None will hit}) = 1 - \frac{1}{20} = \frac{19}{20}$

App III (4) P(either A or B or Both will hit) = ? = Same as part (3)

M-II $P(A \cup B) = ? = P(A) + P(B) - P(A \cap B) = \frac{4}{5} + \frac{3}{4} - \frac{3}{5} = \frac{19}{20}$

(5) Find the prob that target will be hit = ? $= P(\text{at least one will hit}) = \frac{19}{20}$

(6) Find the prob that A hit & B missed = ? $= P(A \cap \bar{B}) = \frac{4}{5} \times \frac{1}{4} = \frac{1}{5}$

(7) " " " " only one will hit = ?

$$P[(A \cap \bar{B}) \cup (\bar{A} \cap B)] = P(E_2 \cup E_3)$$

($A \cap \bar{B}$ → ME ← $\bar{A} \cap B$)

$= P(E_2) + P(E_3)$ ($\because E_2$ & E_3 are ME)

$= P(A \cap \bar{B}) + P(\bar{A} \cap B)$

$= \frac{4}{5} \times \frac{1}{4} + \frac{1}{5} \times \frac{3}{4}$ ($\because$ A & $\bar{B}$ are Ind, similarly $\bar{A}$ & B are Ind)

$= \frac{7}{20}$

(this is the perfect eg to understand how to use the concept of ME & Independency in a single Quest.)

(8) If only one person hit then find the prob that A hit & B missed = ?

Prob III

Condition

original Prob = $P(S) = 1$

Reduced Prob = $P(\text{Condition}) = P(\text{only one person hit}) = \frac{7}{20}$ (using part (7))

fav Prob = $P(A \text{ hit \& } B \text{ missed}) = P(A \cap \bar{B}) = \frac{1}{5} = \frac{1/5}{1}$ (using part 6)

And Prob = $\frac{\text{fav Prob}}{\text{Reduced Prob}} = \frac{1/5}{7/20} = \frac{4}{7}$

Conceptual Questions of Conditional Prob

(just Reduce the S.Space according to condition)

Qs A Couple has 3 kids, If 1st child is known to be a Boy (Condition) then Find the prob that there are exactly 2B in the family.

Sol: original S.Sp = {(BBB), (BBG), (BGB), (BGG), (GBB), (GBG), (GGB), (GGG)} = 8

After Reduced SSp = {1st child is Boy} = {(BBB), (BBG), (BGB), (BGG)} = 4
(Acc to Condition)

fav Cases = {exactly 2B} = {(BBG), (BGB)} = 2

$$\text{Conditional Prob} = \frac{\text{fav Cases}}{\text{R-SSp}} = \frac{2}{4} = \frac{1}{2} \neq \frac{4}{8}$$

App II : Req Prob $= P\left[\frac{\text{exactly 2 B}}{1^{st}\text{ child is Boy}}\right] = P[(B\,B\,G) \text{ or } (B\,G\,B)]$ (ME)

$= 1\times\frac{1}{2}\times\frac{1}{2} + 1\times\frac{1}{2}\times\frac{1}{2} = \frac{1}{2}$

Note: A couple has 3 kids then find the prob that there are exactly 2 B in a family?

Sol: Req Prob $= \frac{\text{fav}}{\text{Total}} = \frac{{}^3C_2}{2^3} = \frac{3}{8}$

& with Cond^n Ans $= \frac{2}{4}$

Q: two Int are to be selected from integers 1, 2, 3, 4 ..... , 11.
If their Sum is Even then Find the prob that Both the selected Integers are odd?

Condition

App I — Lengthy — Do yourself.

App II — 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11

$^{11}C_2 = 55$ pair

2, 4, 6, 8, 10 → $^5C_2 = 10$ pair

1, 3, 5, 7, 9, 11 → $^6C_2 = 15$ pair

Total ways of selecting 2 int = $^{11}C_2 = 55$ pair

Reduced ways : : : : : = {Sum is Even}

= Either (Both are odd) or (Both are Even)

= $^6C_2 + ^5C_2 = 15 + 10 = 25$ Pair

fav Cases = {Both are odd} = $^6C_2 = 15$

and" Prob = $\frac{\text{fav Cases}}{\text{R. Cases}} = \frac{15}{25} = \frac{3}{5}$

Analysis: ① With Cond^n, Req Prob $= P\left(\frac{\text{Both are odd}}{\text{Sum is Even}}\right) = \frac{{}^6C_2}{{}^6C_2 + {}^5C_2} = \frac{15}{15+10} = \frac{15}{25}$

② Had the Cond^n were not these then ans would have been $= ? = \frac{{}^6C_2}{{}^{11}C_2} = \frac{15}{55}$

HW8 App I :- Do yourself

$S = \{(1,1)\ (1,2)\ (1,3)\ (1,4) \ldots (1,10)\ (1,11)$
$(2,1)\ (2,2)\ (2,3)\ (2,4) \ldots (2,10)\ (2,11)$
$(3,4)\ (3,5) \ldots (3,11) \ldots (10,11)\}$

$= {}^{11}C_2 = 55$ pair

THANK - YOU